

"The Indian army is satisfied with our surveillance drones. Now our focus is on exports, as the military drone market is growing fast"

NANDA KUMAR
Founder, Garudan Drones

for watching over forests and for agricultural and administrative purposes. Most importantly, there is accelerating demand from the military for ISR (intelligence, surveillance, reconnaissance) capabilities as well as for machines that can carry out precision strikes on enemy targets. So vital is drone technology that the government's Drone Rules, 2021 sought to streamline and ease operations of drones, also giving a fillip to their manufacture and development in India. The Federation of Indian Chambers of Commerce and Industry (FICCI) has corroborated the move: it says the market potential of Indian drones is projected to be $40 billion in the next 10 years. With the Indian government's avowed intention to make the country a global hub for drone manufacturing, all this has driven a group of entrepreneurs to start ventures for manufacturing drones, where the pull of commercial profit meets the thrill of shaping new technology. In fact, there has been a 34.4 per cent surge in the number of drone start-ups between August 2021 and February this year—from 157 to 221.

Till now, the Indian drone market, especially for defence requirements, has been dominated by big players like Adani Defence and Aerospace (maker of Hermes 450 and Sky Striker, among other drones) and Hindustan Aeronautics Ltd (Heron TP drones), both of whom have tied up with Israeli firms—the biggest supplier of drones to the Indian military. What is unique about start-ups like Jatayu, Bon V Technology, IG Drones, Garudan and CDSpace Robotics is that they are all started by entrepreneurs with an engineering background who are developing their own drone technology, not just manufacturing them here using borrowed foreign technology. Not all have bagged a contract with the Indian army—the common objective—but most of their products are being seriously considered. When it comes to investment, the companies are being supported by the Union and state governments under various policies to help fund local start-ups.

Nanda Kumar is representative of this new crop. The founder of Garudan, which has a wide array of indigenous UAVs, is proud that his three-year-old start-up has struck a deal with a Texas-based company to export mapping drones—the first Indian firm, he claims, to enter the commercial drone market in North America. His next target is to win the Indian army's contract for 750 small drones for surveillance on the northern border. These are to be used by special forces for surveillance within a 5-km radius that can help them execute precision strikes to eliminate high-value targets. Garudan's Vaanvili (a Tamil word meaning 'eye in the sky') series of UAVs cover a range of high-accuracy vehicles suited for long-range to ultra-long-range ISR missions, and hope to meet the 'ruggedness and reliability' standards of the military. In 2018, Nanda Kumar, an electrical engineer from Anna University, Chennai, gathered 12 young engineers and founded Garudan to develop UAVs for diverse surveillance and imaging applications with a combination of rotorcraft, fixed wing craft, customised camera and image processing payloads for medium- to long-range, high accuracy missions.

Entrepreneurs like Nanda Kumar seem to be on to something. A recent Ernst & Young-FICCI report says that the drone sector and its component industries could boost the total value of India's manufacturing to $24 billion by 2030. The Drone Rules, 2021 also emphasises a need for strategic self-reliance in technology. Much of this, of course, pertains to the increasing importance of drones' military uses. According to a report by the global market research firm Allied Market Research, the global military drone market is projected to grow from $11.73 billion